PROPERNESS OF RELATIVE CYCLES SATISFIES H-DESCENT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Functoriality of properness of relative cycles under pullback (relative-cycles chapter, Lemma lemma-proper-functoriality, canonical tag 0H62). cited at proof lines 2137-2138. Base changes of proper cycles are proper, proving the forward direction.
- Support behaviour of families of cycles under base change (relative-cycles chapter, Lemma lemma-support-family, canonical tag 0H4V). cited at proof lines 2144-2146. The support of the pulled back cycle is the inverse image of the support, transferring properness to the closure of the inverse image of the support.
- Functoriality of closed subsets proper over the base (Cohomology of Schemes, Lemma coherent-lemma-functoriality-closed-proper-over-base, canonical tag 0CYQ). cited at proof lines 2150-2152. The image of the closed subset W under the locally closed morphism f is closed and proper over S, containing the support of alpha.

Source reference. Original lemma 0H62: sources/stacks-chow/tags.txt, source lines 21407–21407. Cited at proof lines 2137-2138: forward direction.

Source reference. Original lemma 0H4V: sources/stacks-chow/tags.txt, source lines 21365–21365. Cited at proof lines 2144-2146: support of the pullback is the inverse image of the support.

Source reference. Original lemma 0CYQ: sources/stacks-chow/tags.txt, source lines 16250–16250. Cited at proof lines 2150-2152: the image f(W) is closed and proper over S.

Lemma 0.2. *Let $f : X \to S$ be a morphism of schemes. Assume S locally Noetherian and f locally of finite type. Let $r \geq 0$ be an integer. Let α be a relative r-cycle on X/S. Let $\{g_i : S_i \to S\}$ be a h covering. Then α is proper if and only if each base change $g_i^*\alpha$ is proper.*

Proof. If α is proper, then each $g_i^*\alpha$ is too by Lemma 0H62. Assume each $g_i^*\alpha$ is proper. To prove that α is proper, it clearly suffices to work affine locally on S. Thus we may and do assume that S is affine. Then we can refine our covering

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$\{S_i \to S\}$ by a family $\{T_j \to S\}$ where $g : T \to S$ is a proper surjective morphism and $T = \bigcup T_j$ is an open covering. It follows that $\beta = g^*\alpha$ is proper on $Y = T \times_S X$ over T. By Lemma 0H4V we find that the support of β is the inverse image of the support of α by the morphism $f : Y \to X$. Hence the closure $W \subset Y$ of $f^{-1}\text{Supp}(\alpha)$ is proper over T. Since the morphism $T \to S$ is proper, it follows that W is proper over S. Then by Cohomology of Schemes, Lemma 0CYQ the image $f(W) \subset X$ is a closed subset proper over S. Since $f(W)$ contains $\text{Supp}(\alpha)$ we conclude α is proper. $\square$